Breast Cancer Diagnosis Prediction Using Machine Learning

Student Name: Mohsin
Course: Machine Learning
Instructor: Daniele Bonacorsi & Luca Clissa
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Project Repository

GitHub Project Link:
https://github.com/Mohsin32525/-breast-cancer-ml-project

1. Introduction

Breast cancer is one of the most common and life-threatening diseases worldwide. Early and accurate detection plays a crucial role in improving patient survival rates. Machine learning techniques can assist medical professionals by analyzing patterns in medical data and predicting whether tumors are malignant or benign.

The goal of this project is to apply several machine learning algorithms to the Wisconsin Diagnostic Breast Cancer dataset to classify tumors based on numerical features extracted from cell nuclei images.

2. Dataset Description

The dataset used in this project is the Wisconsin Diagnostic Breast Cancer (WDBC) dataset. It contains measurements computed from digitized images of breast cell nuclei obtained through fine needle aspiration tests.

The dataset contains:

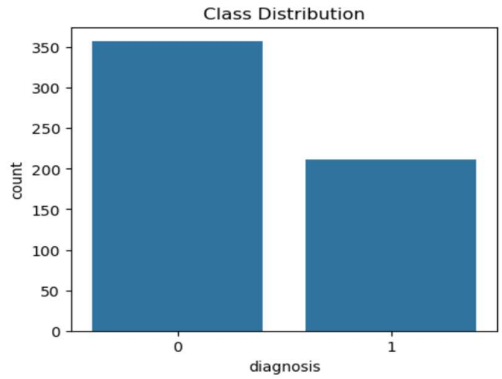
• 569 samples
• 30 numerical features describing cell nucleus characteristics
• Target variable: diagnosis

The diagnosis variable indicates whether a tumor is malignant or benign. The features are divided into three groups:

• Mean measurements
• Standard error measurements
• Worst (largest) measurements

These features describe properties such as radius texture perimeter area smoothness concavity and symmetry.

Figure 1: Class distribution of benign and malignant tumors.



3. Data Preprocessing

Before applying machine learning algorithms several preprocessing steps were performed.

First the diagnosis column was converted from categorical labels (M and B) to numerical values where malignant tumors were encoded as 1 and benign tumors as 0.

Next the ID column was removed since it does not contribute to prediction.

The dataset was then divided into input features (X) and target labels (y). The data was split into training and testing sets using an 80/20 ratio to evaluate model performance.

Finally feature scaling was applied using the StandardScaler method to normalize feature values. Scaling improves the performance of algorithms such as Support Vector Machine and K-Nearest Neighbors.

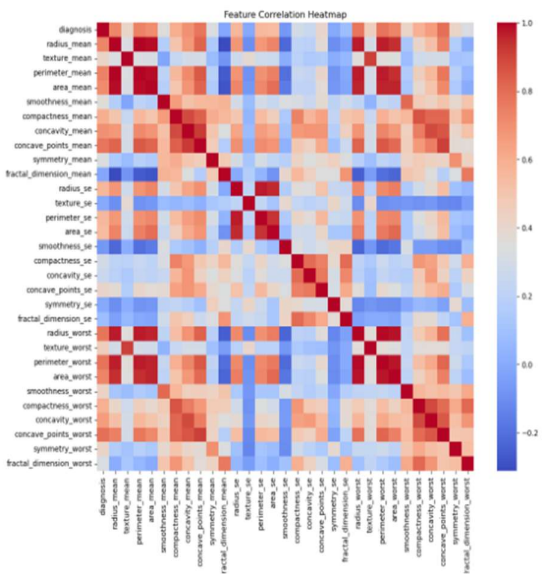


Figure 2: Correlation heatmap of dataset features.

The correlation heatmap shows relationships between the dataset features. Several variables such as radius perimeter and area show strong correlations indicating similar patterns among these measurements.

4. Machine Learning Models

Model	**Accuracy**
Logistic Regression	97%
KNN	94%
Decision Tree	93%
Random Forest	96%
Support Vector Machine	98%

Five machine learning algorithms were trained and evaluated in this project:

- Logistic Regression
- K-Nearest Neighbors
- Decision Tree
- Random Forest
- Support Vector Machine

Each model was trained on the training dataset and evaluated using the testing dataset.

Model performance was measured using classification accuracy. Additional evaluation techniques such as confusion matrix and cross-validation were also applied.

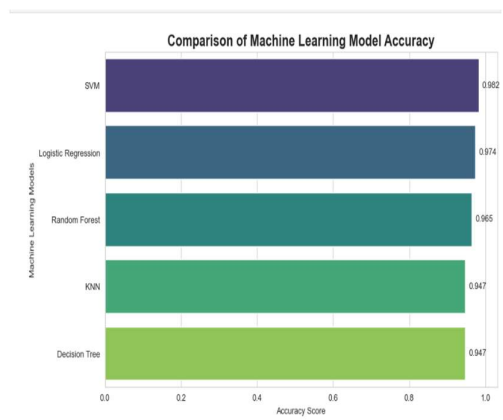


Figure 3: Comparison of model accuracies.

Use the **model comparison bar chart** you created.

Explain:

The results show that the Support Vector Machine model achieved the highest accuracy among all tested models.

5. Results

The results of the machine learning models are summarized below:

The Support Vector Machine model achieved

the best performance with approximately 98% accuracy.

Cross-validation was also performed to confirm that the model performs consistently across different subsets of the dataset.

6. Conclusion

In this project multiple machine learning models were applied to the Wisconsin Diagnostic Breast Cancer dataset to classify tumors as malignant or benign.

The results demonstrate that machine learning algorithms can effectively analyze medical datasets and support early cancer diagnosis. Among the evaluated models Support Vector Machine achieved the highest accuracy.

This project highlights the potential of machine learning in medical decision-support systems.